Sinkhorn revisited - joint calibration to European and One-Touch prices

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Abstract

We apply a Sinkhorn scheme to the problem of constructing an admissible joint law for X_T and the supremum S_T of a continuous martingale X, given target laws μ, ν for X_T, S_T. This corresponds to fitting European and One-Touch prices at all strikes and barrier levels at maturity T. Without the continuity assumption on X, [Rog93] gives an explicit solution to this problem using a "lost-time" martingale with positive-only jumps, but X_T and S_T are co-monotonically coupled which may be less realistic from a financial standpoint (see Section 4 for details). For the continuous case, ν necessarily lies between the classical Azéma-Yor and Perkins bounds, but we show this condition is not sufficient with a simple counterexample. As in [For26], our approach circumvents the need for entropy minimization and duality arguments, and after each Sinkhorn cycle, the law of S_T and the Rogers[Rog93] condition are satisfied exactly, and the X_T-marginal converges to μ (under a finite-entropy feasibility assumption) at rate $1/\sqrt{n}$ in the total variation metric. This can also be solved as an optimal transport linprog problem as in [DR15] where one computes the max or min price of a non-tradable option given tradable options, but the latter also typically leads to a sparse joint law. We show that a (transformed) left-curtain coupling gives an explicit solution to the problem under additional restrictions on ν, and compute a sharp upper price bound for barrier options with supermodular payoffs when the continuity assumption on X is dropped.

Keywords: Sinkhorn algorithm; barrier options; entropic optimal transport; Skorokhod embeddings; excursion theory; left-curtain coupling; joint calibration.

1 Background

The Skorokhod embedding problem (SEP) provides a classical starting point for the problem described in the abstract. The Azéma–Yor[AY79] embedding maximizes the law of the maximum for a given terminal law, and the Perkins[Per86] embedding minimizes the law of the maximum and maximizes the law of the minimum. With an additional intermediate marginal constraint, [BHR01] obtain a sharp bound for the running maximum and construct an attaining martingale. [CO11] derive sharp model-independent bounds for double no-touch options using super- and sub-hedging strategies with European and digital options, and establish optimality using a Perkins embedding for the upper bound and a tilted-Jacka [Jac88] embedding for the lower bound with the tilt parameter K chosen according to the two barriers. [HOST16] extend [BHR01] to the multi-marginal setting, and [COT19] solve the multi-marginal version of the Root SEP.

[Rog93] characterizes the possible joint laws of the maximum and terminal value of a uniformly integrable (UI) continuous martingale, and this result forms the basis of the Sinkhorn scheme in this paper. [Rog93] also characterizes the joint laws for general UI martingales (allowing discontinuities), which is simplified slightly in Lemma 2.3 in [For19] (see the introduction of [For19] for further historical background on the problem considered in this paper).

In a broader context, [BJ16] introduce the left-curtain coupling, a canonical martingale coupling characterized by its left-monotonicity property which minimizes $\mathbb{E}(h(Y-X))$ for costs satisfying $h''' > 0$, such as $h(z) = z^3$, provided the relevant expectations are finite. [HLT16] extend this optimality result to the wider class of payoffs under the Spence–Mirrlees condition $c_{xyy} > 0$, for which the left-curtain coupling is a maximizer, and derive explicit two-branch representations under regularity assumptions, and associated sub/super hedges.

More recently, [BBBL26] describe how the Föllmer martingale is the continuous-time martingale Schrödinger bridge between its own marginals, for which a key component of the proof is just noting that $u_t = \int_0^t \frac{\sigma_s - 1}{1-s} dB_s$ so $\int_0^1 u_t dt = \int_0^1 \int_0^t \frac{\sigma_s - 1}{1-s} dB_s dt = \int_0^1 \frac{\sigma_s - 1}{1-s} (\int_s^1 dt) dB_s = \int_0^1 (\sigma_s - 1) dB_s$, hence (by Itô's isometry and Fubini) $\mathbb{E}(\int_0^1 u_t^2 dt) = \mathbb{E}(\int_0^1 \int_0^t \frac{(\sigma_s-1)^2}{(1-s)^2} ds dt) = \mathbb{E}(\int_0^1 \frac{(\sigma_s-1)^2}{1-s} ds)$, which is the cost function for the continuous-time martingale Schrödinger bridge. The classical (non-martingale) dynamic Schrödinger bridge problem with initial law μ and terminal law ν involves looking for a change of measure of the form $\frac{d\mathbb{P}}{d\mathbb{Q}} = f(X_0)g(X_T)$, where $\mathbb{Q}$ is a Brownian measure with the same initial law; then f and g satisfy two coupled integral equations akin to the Sinkhorn integral equations in e.g. [Guy20], [For26], [JLO23] and the current article.

[Guy20] applies an iterative Sinkhorn scheme to compute an admissible tri-variate law for the SPX-VIX joint calibration problem; [For26] uses an analogous approach for the FX cross-smiles problem, and characterizes the speed of convergence for the marginals under the total variation metric.

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1.1 The Rogers characterization

In the seminal work of [Rog93], the following fundamental result is established [1]:

Theorem 1.1 *(cf. Theorem 3.1 and Lemma 2.3 in [Rog93]; see also Theorem 3.1 in [DR15]). Fix a $T > 0$ and $x_0 \in \mathbb{R}$. A probability measure π on $\{(s,x) : -\infty < x \le s,\ s \ge x_0\}$ is the joint law of (S_T, X_T) (where $S_t = \sup_{0\le u\le t} X_u$) for some continuous martingale X with $X_0 = x_0$ if and only if $\int |x|\pi(ds,dx) < \infty$, $\int x\pi(ds,dx) = x_0$, and*

$$\eta(ds) = s\nu(ds) - \bar{F}_\nu(s)ds \qquad \text{(as measures on } (x_0,\infty)), \tag{1}$$

where $\nu(ds) = \int_{x\le s} \pi(ds,dx)$, $\eta(ds) = \int_{x\le s} x\pi(ds,dx)$, and $\bar{F}_\nu(s) = \nu((s,\infty))$.

Eq (1) has the physical interpretation:

$$\mathbb{P}(S_T > b)db = \mathbb{E}((S_T - X_T)1_{S_T\in db}) = \mathbb{E}((b - X_T)1_{S_T\in db}) \tag{2}$$

and we give a sketch proof of the sufficiency part of the [Rog93] proof in Appendix D which uses basic excursion theory.

In Eq 3.8 in [DR15], the characterization in Theorem 1.1 is re-stated in the more intuitive integrated form: for π supported on $\{x \le s,\ s \ge x_0\}$ with $\int |x|\pi(ds,dx) < \infty$,

$$\int_{\{s\ge a\}} (x-a)\pi(ds,dx) = 0 \qquad \text{for all } a \ge x_0, \tag{3}$$

i.e. $\mathbb{E}(X_T - a;\ S_T \ge a) = 0$ for all $a \ge x_0$, which is clearly a necessary condition from the optional stopping theorem, and the tail condition

$$\lim_{a\to\infty} a\bar{F}_\nu(a) = 0 \tag{4}$$

follows from dominated convergence (see just below Eq 3.8 in [DR15]). Differentiating (3) with respect to a (in the sense of measures) and setting $a = b$ gives Eq (2).

Example 1.1 (Brownian motion stopped at a lower barrier) *(see Remark (ii) after Theorem 3.1 of [Rog93])*[2]*. Let W be a standard Brownian motion with $W_0 = 0$, let $\tau = \inf\{t \ge 0 : W_t = -1\}$, and let $X_t = W_{t\wedge\tau}$. Then X is a continuous martingale on $[0,\infty)$ with $X_\infty = -1$ almost surely, and its supremum $S_\infty = \sup_{t\ge0} X_t$ has law $\nu(ds) = \frac{ds}{(1+s)^2}$ so $\bar{F}_\nu(s) = \nu((s,\infty)) = \frac{1}{1+s}$. The joint law of (S_∞, X_∞) satisfies the local Rogers condition (1) on $(0,\infty)$, but X is not uniformly integrable: $\mathbb{E}(X_\infty) = -1 \ne 0$ and $s\bar{F}_\nu(s) \to 1$ as $s \to \infty$.*

Proof. For $s \ge 0$, the event $\{S_\infty > s\}$ is the event that W reaches s before -1, which has probability $1/(1+s)$ (the gambler's ruin probability, obtained by applying the optional stopping theorem to W stopped at the first exit from $(-1,s)$); this gives $\bar{F}_\nu$ and, on differentiating, the density $\nu(s) = (1+s)^{-2}$. Since $X_\infty = -1$ on every path, the measure $\eta(ds) = \int x\pi(ds,dx)$ is $\eta(ds) = -\nu(ds)$. Hence, for $s > 0$,

$$s\nu(s) - \bar{F}_\nu(s) = \frac{s}{(1+s)^2} - \frac{1}{1+s} = \frac{s-(1+s)}{(1+s)^2} = -\frac{1}{(1+s)^2} = -\nu(s)$$

so $\eta(ds) = s\nu(ds) - \bar{F}_\nu(s)ds$ on $(0,\infty)$, which is Eq (1). On the other hand $\mathbb{E}X_\infty = -1 \ne X_0$, so X is not a UI martingale, and $s\bar{F}_\nu(s) = s/(1+s) \to 1 \ne 0$. ■

2 Outline of problem

Our aim is to construct a continuous martingale X with prescribed marginals μ and ν for X_T and S_T. We refer to (1) as the local Rogers condition, and this condition will form the basis of the Sinkhorn scheme below.

Based on Theorem 1.1 and Eq (4), we impose the following necessary conditions on μ and ν:

Assumption 2.1 *$\int |x|\mu(dx) < \infty$, $\int x\mu(dx) = x_0$ and $\lim_{a\to\infty} a\bar{F}_\nu(a) = 0$, and ν lies between the Perkins and Azéma-Yor (AY) model-independent bounds, i.e.*

$$\sup_{k<b}\{\mu([b,\infty)) + \frac{C_\mu(b) - P_\mu(k)}{b-k}\} \ \le\ \nu([b,\infty)) \ \le\ \inf_{k<b} \frac{C_\mu(k)}{b-k} \tag{5}$$

[3] *for all $b \ge x_0$, where $C_\mu(k) = \int (x-k)^+\mu(dx)$ and $P_\mu(k) = \int (k-x)^+\mu(dx)$ denote the call and put pricing functions.*

[1] See also Remark (i) after Theorem 3.1 in [Rog93]. The results in [Rog93] are formulated for uniformly integrable (UI) martingales on $[0,\infty)$ and their terminal value M_∞. They apply here because a martingale X on $[0,T]$, extended by $X_t = X_T$ for $t \ge T$, is a UI martingale on $[0,\infty)$ with $X_\infty = X_T$ and $\sup_{t\ge0} X_t = S_T$, and conversely a UI continuous martingale can be time-changed onto $[0,T]$ without altering its terminal value or its supremum. Theorem 1.1 is thus a restatement of Theorem 3.1 and Lemma 2.3 in [Rog93], together with Remark (i) there for $x_0 \ne 0$.

[2] See Theorem 4.1 in [Rog93] for another interesting related result

[3] the bounds are sharp here, and attained by the Perkins[Per86] and Azéma-Yor[AY79] Skorokhod embeddings respectively, see also [Obl04],[Cox04],[Hob11] for details.

Definition 2.1 *A probability measure on $\{(s,x) : -\infty < x \le s < \infty,\ s \ge x_0\}$ is called* ***admissible*** *if it has X-marginal μ, S-marginal ν, and satisfies the conditions in Theorem 1.1. We say that the pair (μ,ν) is admissible if an admissible probability measure exists.*

As alluded to in the abstract, the following counterexample shows that ν being a probability measure and lying between the Perkins and Azéma-Yor bounds is not a sufficient condition for an admissible joint law to exist.

Counterexample 2.1 *Let $\mu \sim U(-1,1)$, so $x_0 = 0$, and let ν be the law on $(0,1)$ with tail cdf*

$$\bar{F}_\nu(s) = 1 - s - \tfrac{7}{2}\, s(1-s)^5, \qquad 0 \le s \le 1.$$

The Azéma–Yor and Perkins bounds for μ are $1-s$ and $1-\sqrt{s}$ respectively, and $1-\sqrt{s} \le \bar{F}_\nu(s) \le 1-s$ on $(0,1)$, so ν satisfies Assumption 2.1, but there is no continuous martingale X with $X_T \sim \mu$ and $S_T \sim \nu$.

Proof. Suppose such a martingale exists, and let π be the joint law of (S_T, X_T). By Theorem 1.1, π satisfies (1): $\eta(ds) = s\nu(ds) - \bar{F}_\nu(s)\, ds$ on $(0,\infty)$, where $\eta(ds) = \int_{x\le s} x\, \pi(ds,dx)$. Since $X_T \ge -1$ almost surely, $\eta(ds) \ge -\int_{-1\le x\le s} \pi(ds,dx) = -\nu(ds)$. Substituting the expression for η from Eq (1) into the left-hand side, this reads $s\nu(ds) - \bar{F}_\nu(s)\, ds \ge -\nu(ds)$, i.e.

$$(1+s)\,\nu(ds) \ \ge\ \bar{F}_\nu(s)\, ds \qquad \text{on } (0,1),$$

so $(1+s)\nu(s) \ge \bar{F}_\nu(s)$ for Lebesgue almost every $s \in (0,1)$. But this inequality is violated for $\nu(s) = -\bar{F}'_\nu(s) = 1 + \frac{7}{2}(1-s)^4(1-6s)$ on a neighbourhood of $s = \frac{1}{3}$. ∎

Lemma 2.2 *(Convex-order necessary condition). Suppose an admissible joint law π of (S_T, X_T) exists, and let $\eta(ds) = \int_{x\le s} x\, \pi(ds,dx)$. Then:*
(i) $\eta \ll \nu$, so that $b := d\eta/d\nu$ is well defined ν-almost everywhere and $\mathbb{E}(X_T|S_T) = b(S_T)$; if ν has a density, $b(s) = s - \bar{F}_\nu(s)/\nu(s)$.
(ii) $\mu \succeq_{\mathrm{cx}} \nu \circ b^{-1}$, that is, $\int \phi\, d\mu \ \ge\ \int \phi(b(s))\, \nu(ds)$ for all convex $\phi : \mathbb{R} \to \mathbb{R}$, where $\nu \circ b^{-1}$ is the law of $b(S_T)$.

Proof.
(i) For any Borel set A with $\nu(A) = 0$ we have $\pi(A \times \mathbb{R}) = \mathbb{P}(S_T \in A) = 0$, hence $\eta(A) = \mathbb{E}\big(X_T 1_{\{S_T \in A\}}\big) = 0$; this uses $\mathbb{E}|X_T| < \infty$. Thus $\eta \ll \nu$, and the Radon–Nikodym derivative $b = d\eta/d\nu$ exists. By definition of η, $\int_A b\, d\nu = \eta(A) = \mathbb{E}\big(X_T 1_{\{S_T \in A\}}\big)$ for every Borel A, which is the statement $\mathbb{E}(X_T \,|\, S_T) = b(S_T)$. When ν has a density, the local Rogers condition (1) gives $\eta(ds) = (s\nu(s) - \bar{F}_\nu(s))ds$ and hence the formula for b.

(ii) For convex ϕ, the tower property and the conditional Jensen inequality give

$$\int \phi\, d\mu = \mathbb{E}\,\phi(X_T) = \mathbb{E}\big[\mathbb{E}(\phi(X_T)\,|\,S_T)\big] \ \ge\ \mathbb{E}\,\phi\big(\mathbb{E}(X_T\,|\,S_T)\big) = \mathbb{E}\,\phi(b(S_T)) = \int \phi(b(s))\nu(ds).$$

∎

Proposition 2.3 (Sufficiency when b is increasing) *Suppose Assumption 2.1 holds, that the support of ν is an interval, that ν has a density which is positive and continuously differentiable on the interior of its support, and that $s \mapsto 1/\bar{F}_\nu(s)$ is strictly convex there. Then $b(s) = s - \bar{F}_\nu(s)/\nu(s)$ is strictly increasing there. Let $Z = b(S_T)$ and let ν_Z denote the law of Z.*

(i) If $\mu \succeq_{\mathrm{cx}} \nu_Z$, then there exists a martingale coupling $m(dz,dx) = \nu_Z(dz)m_z(dx)$ of ν_Z and μ, i.e. $\int x m_z(dx) = z$ for ν_Z-almost every z. For any such m, the joint law

$$\pi(ds,dx) = \nu(ds)\, m_{b(s)}(dx)$$

of (S_T, X_T) — draw $S_T \sim \nu$ and then X_T from the kernel of m at the point $z = b(S_T)$ — has marginals ν and μ and satisfies the local Rogers condition (1); π is admissible if and only if m_z is supported on $(-\infty, b^{-1}(z)]$ for ν_Z-almost every z.

(ii) Assume that $\mu \succeq_{\mathrm{cx}} \nu_Z$, so the left-curtain coupling of (ν_Z, μ) exists by [BJ16], and in addition that μ has a density, that the pair (ν_Z, μ) satisfies the regularity conditions of [HLT16] (Assumptions 3.3 and 3.6 there), and that the densities satisfy the following single-crossing condition: writing $z_0 = b(x_0)$ and r_{ν_Z} for the left and right endpoints of the support of ν_Z, there exists $z_* \in (z_0, r_{\nu_Z})$ such that $f_{\nu_Z} > f_\mu$ on (z_0, z_*) and $f_{\nu_Z} < f_\mu$ on (z_*, r_{ν_Z}). This ensures that the left-curtain coupling of (ν_Z, μ) has no identity part, as in Example 2.1. If the left-curtain coupling, whose branches $x_-(s) \le x_+(s)$ are determined by

$$\int_{x_-(s)}^{x_+(s)} \mu(dx) = 1 - \bar{F}_\nu(s), \qquad \int_{x_-(s)}^{x_+(s)} x\mu(dx) = \int_{x_0}^{s} b(u)\nu(u)du,$$

satisfies $x_+(s) \le s$ for ν-almost every s, then (μ,ν) is admissible, an admissible law being

$$\pi(dx|s) = q(s)\delta_{x_+(s)} + (1-q(s))\,\delta_{x_-(s)}, \qquad q(s) = \frac{b(s) - x_-(s)}{x_+(s) - x_-(s)}.$$

Proof. See Appendix E. ■

Example 2.1 (An explicit family) *Let $\mu \sim U(-1,1)$, so $x_0 = 0$, and let ν have tail $\bar{F}_\nu(s) = (1-s)(1-\theta s)$ on $[0,1]$ with $0 < \theta \leq \frac{1}{2}$, so that $\nu(s) = 1+\theta-2\theta s$. Then $\bar{F}_\nu$ lies between the Azéma-Yor bound $1-s$ and the Perkins bound $1-\sqrt{s}$, and is tangent to the latter at $s=1$ when $\theta = \frac{1}{2}$. Since $\log(1/\bar{F}_\nu) = -\log(1-s) - \log(1-\theta s)$ is convex, $1/\bar{F}_\nu$ is strictly convex and, by Proposition 2.3, $b(s) = s - \frac{(1-s)(1-\theta s)}{1+\theta-2\theta s}$ is strictly increasing, from $b(0) = -1/(1+\theta)$ to $b(1) = 1$.*

The single-crossing condition in Proposition 2.3 holds here. Indeed, $f_{\nu_Z}(b(s)) = \nu(s)/b'(s)$ is strictly decreasing for $0 < s < 1$ when $0 < \theta \leq 1/2$, with $f_{\nu_Z}(b(0)+) = (1+\theta)^3/(2(1+\theta+\theta^2)) > 1/2$ and $f_{\nu_Z}(b(1)-) = (1-\theta)/2 < 1/2$. Since $f_\mu = 1/2$ on $(-1,1)$, the densities cross exactly once, from above to below. All the quantities in Proposition 2.3 are then explicit. Since μ has density $\frac{1}{2}$, the two equations for the branches are $\frac{1}{2}(x_+ - x_-) = 1 - \bar{F}_\nu(s)$ and $\frac{1}{4}(x_+^2 - x_-^2) = \int_0^s b(u)\nu(u)du$.

The branches are monotone, x_- decreasing from $b(0)$ to -1 and x_+ increasing from $b(0)$ to 1, and the support condition $x_+(s) \leq s$ reduces to $2\theta s + \theta^2 s(1-s) \leq 1$, whose left-hand side is increasing in s and equals 2θ at $s=1$: it holds on all of $[0,1]$ precisely when $\theta \leq \frac{1}{2}$. Hence, provided $\mu \succeq_{\text{cx}} \nu_Z$ as required in Proposition 2.3(ii), for every $\theta \in (0, \frac{1}{2}]$ the two-point law $\pi(dx|s) = q(s)\delta_{x_+(s)} + (1-q(s))\delta_{x_-(s)}$ is an explicit admissible joint law of (S_T, X_T), and the pair (μ,ν) is admissible. The threshold $\theta = \frac{1}{2}$ is the same as the one at which $\bar{F}_\nu$ ceases to dominate the Perkins bound at $s=1$.

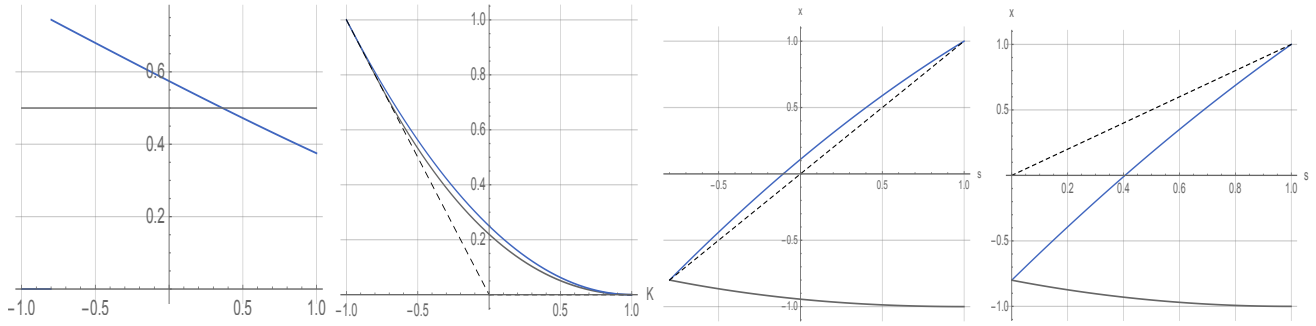


Figure 1: In the 1st plot we plot the densities of Z and X for $\theta = \frac{1}{4}$, and in the 2nd plot we check the convex order condition for Example 2.1 by pricing call options on Z for a range of strike values K (solid grey) and X (blue). In the 3rd plot we see the upper and lower branches $T_u(z)$ (blue) and $T_d(z)$ (solid grey) for the left-curtain coupling, and on the right we see the corresponding $x_+(s)$ (blue) and $x_-(s)$ (solid grey), and we (correctly) see that $x_+(s) \leq s$.

Motivated by financial applications, we henceforth assume μ only has support on the positive real line. If X is an asset price and interest rates and dividends are zero, μ can be inferred from European option prices at all strikes via the Breeden-Litzenberger formula (using e.g. an SVI smile interpolation in practice, as in Figure 8 below), and $\nu([B,\infty))$ is the price of a One-Touch option on X with barrier level $B > X_0$. Once we have an admissible joint law for X_T and S_T, we can then price other barrier options whose payoffs depend jointly on X_T and S_T, which is the primary application here. We also remark that barrier options are most actively traded in FX options, and are predominantly over-the-counter instruments.

2.1 Sinkhorn integral equations

From here on, we further assume that μ, ν have densities with respect to the Lebesgue measure, and set

$$r(s) = s\nu(s) - \bar{F}_\nu(s)$$

i.e. the density on the right-hand side of (1).

Let $p(s,x)$ be a candidate joint density for (S,X). Based on the $\mathbb{E}(X_T 1_{S_T \in db})$ term in (2) and the entropic optimal transport literature, we make the exponential ansatz:

$$p(s,x) = R(s,x)e^{u(x)+v(s)+x\lambda(s)}$$

where R is a reference density (see Assumption 2.4 for conditions on R). Then the X-marginal condition is

$$\mu(x) = e^{u(x)} \int_x^\infty R(s,x)e^{v(s)+\lambda(s)x} ds$$

which we can rearrange as

$$u(x) = \log \mu(x) - \log\Big(\int_x^\infty R(s,x)e^{v(s)+\lambda(s)x} ds\Big) \tag{6}$$

if the integral term lies in $(0,\infty)$ (this will be addressed in Lemma C.1). Similarly, the S-marginal condition is

$$\nu(s) = e^{v(s)} \int_0^s R(s,x) e^{u(x)+\lambda(s)x} dx$$

and the local Rogers condition in (1) is

$$r(s) = e^{v(s)} \int_0^s xR(s,x) e^{u(x)+\lambda(s)x} dx.$$

Dividing by the S-marginal equation then gives

$$\frac{\int_0^s xR(s,x)e^{u(x)+\lambda(s)x}dx}{\int_0^s R(s,x)e^{u(x)+\lambda(s)x}dx} = b(s) = \frac{r(s)}{\nu(s)} \tag{7}$$

where $b(s)$ is defined for ν-almost every s. Under suitable technical conditions (see Lemma 2.5 for details), $\lambda(s)$ can be computed from (7) as a one-dimensional root-finding exercise. The S-marginal equation gives

$$v(s) = \log \nu(s) - \log\Big(\int_0^s R(s,x)e^{u(x)+\lambda(s)x}dx\Big). \tag{8}$$

Hence we see that (6), (7) and (8) are the three coupled equations for u, λ and v. [Guy20] and [For26] employ a similar approach for the SPX-VIX and FX cross-smile calibration problems.

2.2 The Sinkhorn scheme

The coupled equations above suggest the following iterative scheme. Set $u_0 = v_0 = \lambda_0 = 0$. Given (u_n, v_n, λ_n), first set

$$u_{n+1}(x) = \log \mu(x) - \log\Big(\int_x^\infty R(s,x)e^{v_n(s)+\lambda_n(s)x}ds\Big). \tag{9}$$

This enforces the X-marginal exactly. Next, choose $\lambda_{n+1}(s)$ as the solution of

$$\frac{\int_0^s xR(s,x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x}dx}{\int_0^s R(s,x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x}dx} = b(s). \tag{10}$$

This is a scalar root-finding problem in λ for each s; existence and uniqueness of the root for ν-almost every s are established in Lemma 2.5. Finally set

$$v_{n+1}(s) = \log \nu(s) - \log\Big(\int_0^s R(s,x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x}dx\Big). \tag{11}$$

The next u-update restores the X-marginal, which is the only constraint disturbed by the completed (λ, v) update. In practice, one iteration therefore consists of an explicit u-update, one scalar root-find for $\lambda_{n+1}(s)$ for ν-almost every s, and an explicit v-update.

For probability measures P and Q we write $H(P|Q) = \int \log \frac{dP}{dQ}\, dP$ if $P \ll Q$ and $H(P|Q) = +\infty$ otherwise. The integral is well defined in $[0,\infty]$: writing $\rho = dP/dQ$, $\int (\log\rho)^- dP = \int \rho(\log\rho)^- dQ \le 1/e$, since $\rho\log\rho \ge -1/e$ for $\rho \ge 0$, and $H(P|Q) \ge 0$ by Jensen's inequality. [4]

From here on we make the following further assumption:

Assumption 2.4 *$\mu = 0$ outside $0 \le x \le K$, $\nu = 0$ outside $X_0 \le s \le K$ for some $K < \infty$, and $\mu(x) > 0$ for Lebesgue almost every $x \in (0,K)$ and $\nu(s) > 0$ for Lebesgue almost every $s \in (X_0, K)$. The reference density R is a probability density supported on*

$$s \ge X_0, \qquad 0 < x \le s$$

with $R(s,x) > 0$ for every $s \ge X_0$ and Lebesgue almost every $x \in (0,s)$. Finally, there exists an admissible probability density p^* such that $H(p^*|R) < \infty$.

Since $H(p^*|R) < \infty$, we have $p^* \ll R$, so $0 < X \le S$ p^*-almost surely. Since p^* is a density, $p^*(X = S) = 0$. Hence, using $b(s) = \mathbb{E}_{p^*}(X|S=s)$, (by the local Rogers condition), we obtain

$$0 < b(s) < s \tag{12}$$

for ν-almost every s.

[4] The same bound justifies evaluating $H(P|Q)$ as a double integral computed one variable at a time whenever P and Q are given by densities.

Remark 2.1 (Reference density for Geometric Brownian motion) A natural choice for the reference density R is the joint law of (S_T, X_T) for driftless geometric Brownian motion $X_t = X_0 e^{\sigma W_t - \sigma^2 t/2}$. $Y_t = \log X_t$ is a Brownian motion with drift $-\sigma^2/2$ and volatility σ started at $\log X_0$, and the reflection principle and Girsanov's theorem give the joint density $f_{M_T,Y_T}(b,y)$ of $M_T = \max_{t\le T} Y_t$ and Y_T. Setting $s = e^b$, $x = e^y$ we find that

$$R(s,x) = \frac{2\,\ell(s,x)}{\sigma^3\sqrt{2\pi T^3}\; s\,x}\sqrt{\frac{X_0}{x}}\; e^{-\frac{\ell(s,x)^2}{2\sigma^2 T} - \frac{\sigma^2 T}{8}}, \qquad \ell(s,x) = \log\frac{s^2}{xX_0}, \tag{13}$$

for $x > 0$ and $s \ge \max(x, X_0)$, and $R(s,x) = 0$ otherwise. The tail of its S-marginal is

$$\bar F_R(B) = \int_B^\infty \int_0^s R(s,x)\,dx\,ds = \Phi\Big(\frac{-\log(B/X_0) - \sigma^2 T/2}{\sigma\sqrt{T}}\Big) + \frac{X_0}{B}\,\Phi\Big(\frac{-\log(B/X_0) + \sigma^2 T/2}{\sigma\sqrt{T}}\Big), \qquad B \ge X_0,$$

where Φ is the standard Normal distribution function. Since geometric Brownian motion is a continuous martingale, R satisfies the local Rogers condition (1) exactly, and it is positive on the interior of its support, as required in Assumption 2.4. Note however that R is not supported on $s \le K$, so the initial iterate $p_{0,0,0} = R$ (since $u_0 = v_0 = \lambda_0 = 0$) charges the regions $\{s > K\}$ and $\{x > K\}$; this is why several statements in Section 3 and Appendix C are restricted to $n \ge 1$.

Lemma 2.5 *Suppose Assumption 2.4 holds. Let u be measurable and finite Lebesgue almost everywhere on $(0,K)$ and suppose that, for ν-almost every s, there exists at least one finite value of λ for which $0 < \int_0^s R(s,x)e^{u(x)+\lambda x}dx < \infty$. Then for ν-almost every s there exists a unique finite $\lambda(s)$ such that*

$$\frac{\int_0^s xR(s,x)e^{u(x)+\lambda(s)x}dx}{\int_0^s R(s,x)e^{u(x)+\lambda(s)x}dx} = b(s). \tag{14}$$

Moreover, λ admits a measurable version, and for ν-almost every s, the integral $\int_0^s R(s,x)e^{u(x)+\lambda x}dx$ is positive and finite for every finite λ, and in particular for $\lambda = \lambda(s)$.

Proof. See Appendix A. ■

3 Convergence for the Sinkhorn scheme

For the convergence argument, define $p_{i,j,k}(s,x) = R(s,x)e^{u_i(x)+v_j(s)+\lambda_k(s)x}$. Thus one complete cycle is $p_{n,n,n} \longrightarrow p_{n+1,n,n} \longrightarrow p_{n+1,n,n+1} \longrightarrow p_{n+1,n+1,n+1}$, corresponding to the u-update, the λ-update and the v-update respectively. By Eq (9), $(p_{n+1,n,n})^X = \mu$, so $p_{n+1,n,n}$ is a probability density. By Eq (11), $(p_{n+1,n+1,n+1})^S = \nu$, so every completed iterate is also a probability density. Moreover, Eq (10) gives

$$\frac{\int_0^s xp_{n+1,n+1,n+1}(s,x)dx}{\int_0^s p_{n+1,n+1,n+1}(s,x)dx} = b(s).$$

Since the denominator is $\nu(s)$ by construction, we see that $\int_0^s xp_{n+1,n+1,n+1}(s,x)dx = b(s)\nu(s) = r(s)$. Hence every completed iterate $p_{n,n,n}$ with $n \ge 1$ not only has the correct S-marginal but satisfies the local Rogers condition. The intermediate density $p_{n+1,n,n+1}$ need not have mass one, so in the entropy calculation below the λ- and v-updates are treated together.

Lemma 3.1 *For every $n \ge 0$ for which the u-update is well defined,*

$$\frac{p_{n+1,n,n}(s,x)}{p_{n,n,n}(s,x)} = \frac{\mu(x)}{(p_{n,n,n})^X(x)} \tag{15}$$

and

$$H(p_{n+1,n,n}|p_{n,n,n}) = H(\mu|(p_{n,n,n})^X). \tag{16}$$

Proof. Since $p_{n+1,n,n}$ and $p_{n,n,n}$ differ only in the u-potential, $p_{n+1,n,n}(s,x)/p_{n,n,n}(s,x) = e^{u_{n+1}(x)-u_n(x)}$. Integrating $p_{n,n,n}(s,x) = R(s,x)e^{u_n(x)+v_n(s)+\lambda_n(s)x}$ over s gives $(p_{n,n,n})^X(x) = e^{u_n(x)}\int_x^\infty R(s,x)e^{v_n(s)+\lambda_n(s)x}ds$, so Eq (9) can be written as $e^{u_{n+1}(x)} = \mu(x)e^{u_n(x)}/(p_{n,n,n})^X(x)$, which gives Eq (15). Hence, since $(p_{n+1,n,n})^X = \mu$,

$$H(p_{n+1,n,n}|p_{n,n,n}) = \int\int p_{n+1,n,n}(s,x)\log\frac{\mu(x)}{(p_{n,n,n})^X(x)}dxds = \int \mu(x)\log\frac{\mu(x)}{(p_{n,n,n})^X(x)}dx = H(\mu|(p_{n,n,n})^X),$$

which is Eq (16). ■

Proposition 3.2 *Suppose Assumption 2.4 holds. Then $\|(p_{n,n,n})^X - \mu\|_{\mathrm{TV}} \longrightarrow 0$ as $n \to \infty$.* [5]

[5] see section 1.1 in [Nutz22] for the definition of the total variation distance $\|\cdot\|_{\mathrm{TV}}$ here

Proof. By Lemma C.1, every step of the scheme is well defined and $H(p^*|p_{n,n,n})$ and $H(p^*|p_{n+1,n,n})$ are finite for every $n \geq 0$. Adding Eqs (C-1) and (C-2), the term $H(p^*|p_{n+1,n,n})$ cancels and we obtain, for every $n \geq 0$,

$$H(p_{n+1,n,n}|p_{n,n,n}) + H(p_{n+1,n+1,n+1}|p_{n+1,n,n}) = H(p^*|p_{n,n,n}) - H(p^*|p_{n+1,n+1,n+1}). \tag{17}$$

Now sum Eq (17) over $n = 0, 1, \ldots, N-1$. On the right-hand side the sum telescopes, since the second term for index n cancels the first term for index $n+1$, leaving only the first term for $n = 0$ and the second term for $n = N-1$:

$$\begin{aligned}\sum_{n=0}^{N-1} (H(p_{n+1,n,n}|p_{n,n,n}) + H(p_{n+1,n+1,n+1}|p_{n+1,n,n})) &= \sum_{n=0}^{N-1} (H(p^*|p_{n,n,n}) - H(p^*|p_{n+1,n+1,n+1})) \\ &= H(p^*|p_{0,0,0}) - H(p^*|p_{N,N,N}) \\ &= H(p^*|R) - H(p^*|p_{N,N,N}) \leq H(p^*|R),\end{aligned}$$

since $p_{0,0,0} = R$ and entropy is non-negative. Letting $N \to \infty$ gives

$$\sum_{n=0}^{\infty} \big(H(p_{n+1,n,n}|p_{n,n,n}) + H(p_{n+1,n+1,n+1}|p_{n+1,n,n})\big) \leq H(p^*|R) \tag{18}$$

so in particular, $H(p_{n+1,n,n}|p_{n,n,n}) \longrightarrow 0$. By Lemma 3.1, $H(p_{n+1,n,n}|p_{n,n,n}) = H(\mu|(p_{n,n,n})^X)$. Therefore $H(\mu|(p_{n,n,n})^X) \longrightarrow 0$ and Pinsker's inequality gives $\|(p_{n,n,n})^X - \mu\|_{\mathrm{TV}} \longrightarrow 0$. ■

3.1 Rate of convergence

Proposition 3.3 *Under the assumptions of Proposition 3.2, for every $n \geq 1$, $H((p_{n,n,n})^X|\mu) \leq \frac{H(p^*|R)}{n}$. Hence $\|(p_{n,n,n})^X - \mu\|_{\mathrm{TV}} \leq \sqrt{\frac{2H(p^*|R)}{n}}$, and $(p_{n,n,n})^X \to \mu$ in total variation at rate $O(n^{-1/2})$.*

Proof. Write $q_n = p_{n,n,n}$. Since $(p_{n+1,n,n})^X = \mu$, the Data Processing inequality (DPI, cf. Lemma 1.6 in [Nutz22]) gives

$$H(q_{n+1}^X|\mu) \leq H(q_{n+1}|p_{n+1,n,n}). \tag{19}$$

From Eq (15) $p_{n+1,n,n}(s,x)/q_n(s,x) = \mu(x)/q_n^X(x)$ for $n \geq 1$, so $q_n(s,x)/p_{n+1,n,n}(s,x) = q_n^X(x)/\mu(x)$ q_n-almost everywhere, since for $n \geq 1$ the density q_n^X is supported in $(0,K)$, where $\mu > 0$ almost everywhere. Hence

$$H(q_n|p_{n+1,n,n}) = \int \log \frac{q_n(s,x)}{p_{n+1,n,n}(s,x)} q_n(s,x)dsdx = \int \log \frac{q_n^X(x)}{\mu(x)} q_n(s,x)dsdx = \int \log \frac{q_n^X(x)}{\mu(x)} q_n^X(x)dx = H(q_n^X|\mu). \tag{20}$$

In particular, by Eq (19) with $n-1$ in place of n and Eq (18), $H(q_n|p_{n+1,n,n}) = H(q_n^X|\mu) \leq H(q_n|p_{n,n-1,n-1}) \leq H(p^*|R) < \infty$ for $n \geq 1$.

For $n \geq 1$, both q_n and q_{n+1} have S-marginal ν and conditional mean $b(s)$ by Lemma C.1. Moreover, as shown in the proof of Lemma C.1, $q = p_{n+1,n,n}$ satisfies the hypotheses of Lemma B.1 in Appendix B, with $\lambda(s) = \lambda_{n+1}(s) - \lambda_n(s)$ and with $q' = p_{n+1,n+1,n+1} = q_{n+1}$. Hence Lemma B.1, with $p = q_n$, gives

$$H(q_n|p_{n+1,n,n}) = H(q_n|q_{n+1}) + H(q_{n+1}|p_{n+1,n,n}). \tag{21}$$

By Eq (21), $H(q_{n+1}|p_{n+1,n,n}) \leq H(q_n|p_{n+1,n,n})$. Combining this with Eqs (19) and (20) gives

$$H(q_{n+1}^X|\mu) \leq H(q_{n+1}|p_{n+1,n,n}) \leq H(q_n|p_{n+1,n,n}) = H(q_n^X|\mu).$$

On the other hand, Eqs (19) and (18) give

$$\sum_{n=1}^{\infty} H(q_n^X|\mu) \leq \sum_{n=0}^{\infty} H(q_{n+1}|p_{n+1,n,n}) \leq H(p^*|R).$$

Since $H(q_n^X|\mu)$ is non-increasing and non-negative, $nH(q_n^X|\mu) \leq \sum_{k=1}^n H(q_k^X|\mu) \leq H(p^*|R)$, which proves the first claim. Finally, Pinsker's inequality gives $\|q_n^X - \mu\|_{\mathrm{TV}} \leq \sqrt{2H(q_n^X|\mu)} \leq \sqrt{\frac{2H(p^*|R)}{n}}$. ■

4 Non-continuous martingales and extremal prices/hedges for general barrier options

If we relax the condition that X is continuous, the upper bound in (5) remains unchanged but the Perkins bound is replaced by the trivial lower bound $\mathbb{P}^{X\sim\mu}(\max(X, X_0) \geq b)$ (see e.g. the introduction of [For19] for details). In

this case there is a simple explicit solution to the problem addressed in this paper described at the end of page 459 and page 460 in [Rog93]: let W be a Brownian motion with $W_0 = x_0$ and set $A_t \equiv \int_0^t 1_{\{W_u < c^{-1}(\bar{W}_u)\}} du$, $\tau_t \equiv \inf\{u : A_u > t\}$ and $M_t = W_{\tau_t \wedge \tau_{\text{AY}}}$ where $\bar{W}_t = \sup_{0 \le u \le t} W_u$ and $\tau_{\text{AY}} = \inf\{u : W_u < b_\mu^{-1}(\bar{W}_u)\}$ with $b_\mu(x) = \mathbb{E}^{X \sim \mu}(X | X \ge x)$ so τ_{AY} is the usual AY stopping time, and $M_\infty = W_{\tau_{\text{AY}}} = b_\mu^{-1}(\bar{W}_{\tau_{\text{AY}}})$ and $\bar{M}_\infty = c^{-1}(\bar{W}_{\tau_{\text{AY}}})$ with $c(s) = b_\mu(F_\mu^{-1}(F_\nu(s)))$ and $c^{-1}(y) = \inf\{s \ge x_0 : c(s) \ge y\}$. F_μ, F_ν denote the respective CDFs here, and $x \le c(x) \le b_\mu(x)$ for $x \ge x_0$ so M_∞ and $\bar{M}_\infty$ are linked via the simple co-monotone coupling and M has positive-only jumps since c^{-1} is non-decreasing (see Fig 1 in [Rog93]). Note there is no need for a Sinkhorn scheme here since we have an explicit (singular) solution, unless one seeks a joint density for the terminal value and supremum.

Consider a payoff g such that $|g(s,x)| \le f(s) + h(x)$ for some $f \in L^1(\nu)$ and $h \in L^1(\mu)$, so $\mathbb{E}(|g(S,X)|) < \infty$ for every coupling of ν and μ. Then (by the rearrangement inequality) M maximises $\mathbb{E}(g(\bar{M}_\infty, M_\infty))$ over all martingales M with $M_\infty \sim \mu$ and $\bar{M}_\infty \sim \nu$ whenever g is *supermodular*, i.e. for every $x_1 < x_2$ the increment $s \mapsto g(s,x_2) - g(s,x_1)$ is non-decreasing in s; this holds for e.g. the up-and-in call payoff $g(s,x) = 1_{\{s \ge B\}}(x-K)^+$, whose increment is $1_{\{s \ge B\}}((x_2 - K)^+ - (x_1 - K)^+)$, zero for $s < B$ and a non-negative constant for $s \ge B$.

To see this in terms of a superhedge, note that for arbitrary s, x, if $a \ge K$,

$$1_{\{s \ge B\}}(x-K)^+ \le (x-a)^+ + (a-K)1_{\{s \ge B\}}.$$

Indeed, if $s < B$ the left-hand side is zero. If $s \ge B$, the inequality reduces to $(x-K)^+ \le (x-a)^+ + (a-K)$, which follows from $(u+v)^+ \le u^+ + v^+$ with $u = x - a$ and $v = a - K$.

For the co-monotone coupling this inequality becomes an equality. Assuming μ has a density as above, choose a so that $\mu((a,\infty)) = \nu((B,\infty))$. Then under the co-monotone coupling, $\{x \ge a\} = \{s \ge B\}$. Suppose first that $a \ge K$. If $s < B$, then $x < a$, so $(x-a)^+ = 0$ and both sides are zero. If $s \ge B$, then $x \ge a \ge K$, so $(x-a)^+ + (a-K) = x - a + a - K = x - K = (x-K)^+$. Hence the superhedge is attained pathwise by the co-monotone coupling.

Finally, if $a < K$, then for arbitrary s, x, $1_{\{s \ge B\}}(x-K)^+ \le (x-K)^+$, so the European call itself is a superhedge. Under the co-monotone coupling, whenever $(x-K)^+ > 0$ we have $x > K > a$. Hence $x \ge a$ and $s \ge B$. Thus $1_{\{s \ge B\}}(x-K)^+ = (x-K)^+$, so the vanilla-call superhedge is again attained pathwise by the co-monotone coupling.

5 Numerical results

Figures 1–8 illustrate the left-curtain construction, numerical Sinkhorn calibration results, and simulations of Rogers' lost-time martingale. Note the numerical examples with $\mu = U(-1,1)$ distributions use integration over $\{s \ge 0, -1 \le x \le s \le 1\}$.

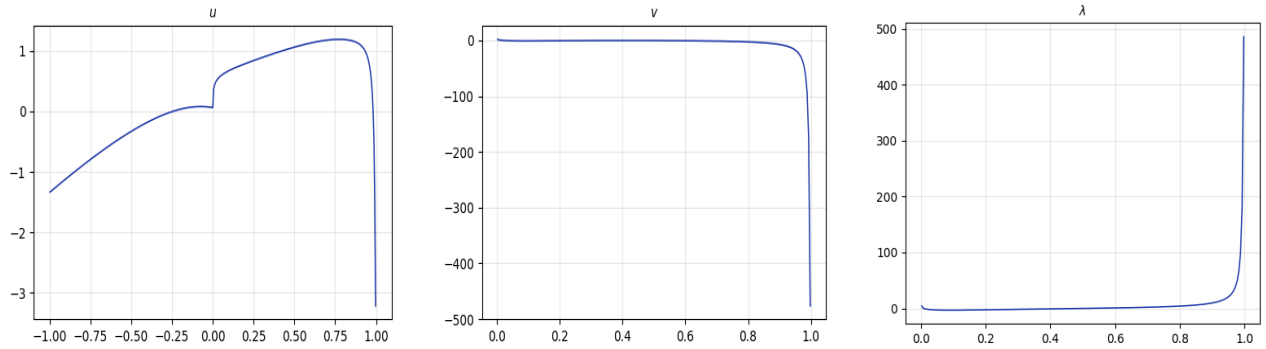


Figure 2: Here we have plotted u, v, λ for a target law ν which is a $(\frac{3}{5}, \frac{2}{5})$ weighting of the AY and Perkins laws respectively (so trivially an admissible (albeit singular) joint law exists) with $\mu = U(-1, 1)$. We take R to be the (re-normalized) product measure $\mu \otimes \nu$ on the region $s \geq x \vee 0$, and we use 400 (equidistant) steps for the X direction and 200 for S.

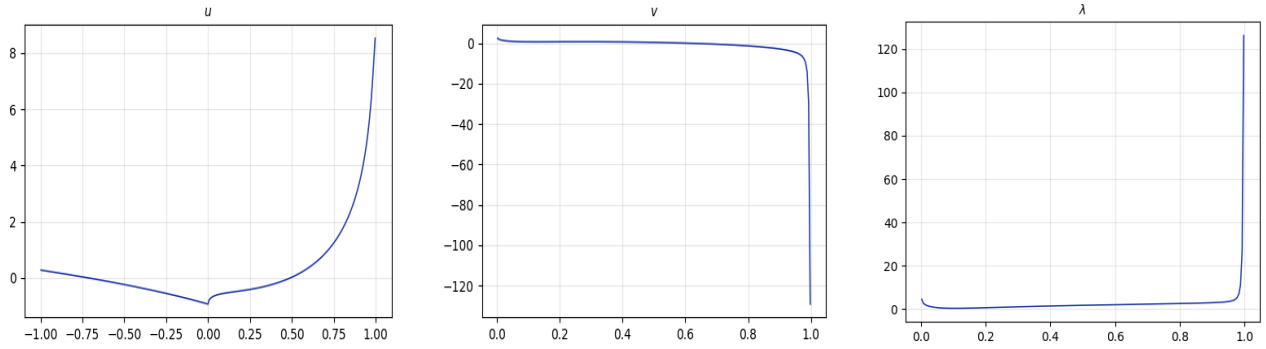


Figure 3: Here we have plotted u, v, λ for a target law ν with cdf $\nu([0, s]) = s^p$ for $s \in [0, 1]$ with $p = 3/4$, with all other parameters unchanged.

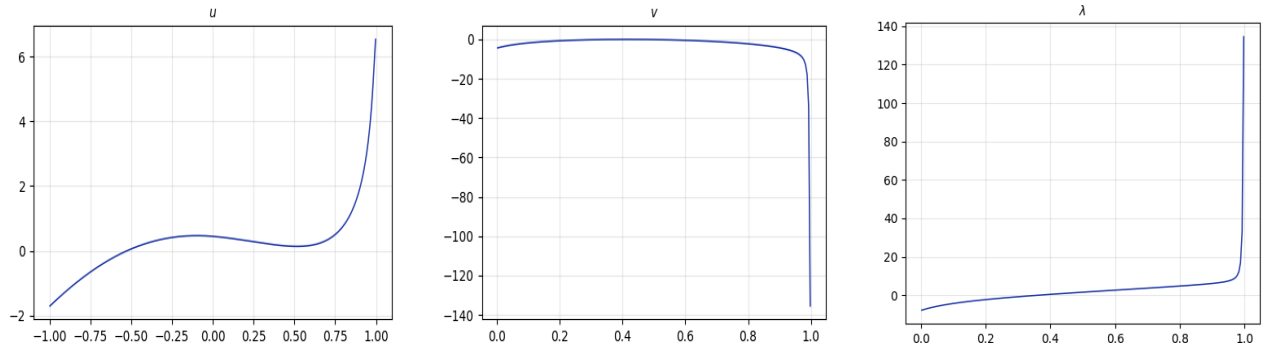


Figure 4: Here we have plotted u, v, λ for a target law ν with cdf $\nu([0,s]) = 1-(1-s)(1-\theta s)$ for $s \in [0,1]$ (see Example 2.1) with $\theta = 1/4$, with all other parameters unchanged.

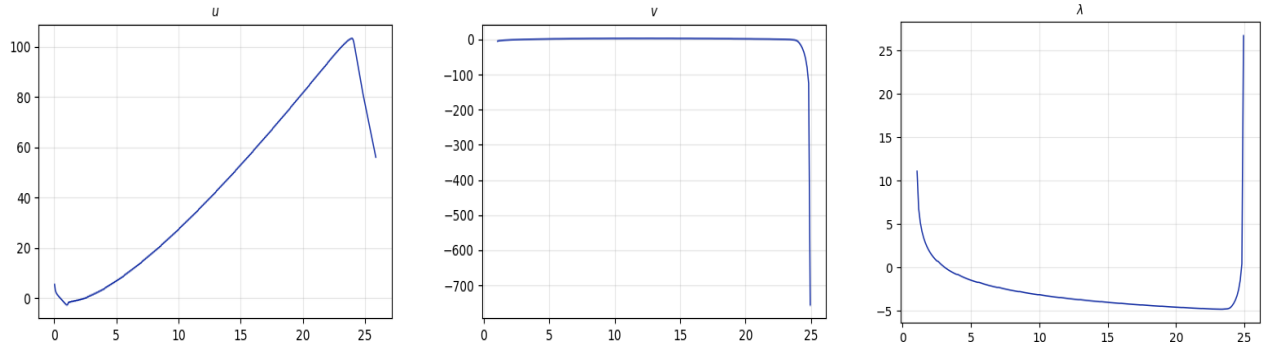


Figure 5: Here we have plotted u, v, λ for a target law ν which is a $(\frac{1}{3}, \frac{2}{3})$ weighting of AY and Perkins with $\mu = \text{Exp}(1)$ and R given by (13) with $\sigma = T = 1$. In this case the AY optimal strike in (5) is $k^*(b) = b - 1$ and the Perkins optimal strike is $k_*(b) = -1 + b - W((be^b - 1 - e^b)/e)$, where $W(.)$ denotes the principal branch of the Lambert-W function, i.e. the unique solution $w \geq -1$ of $we^w = z$ for $z \geq -1/e$

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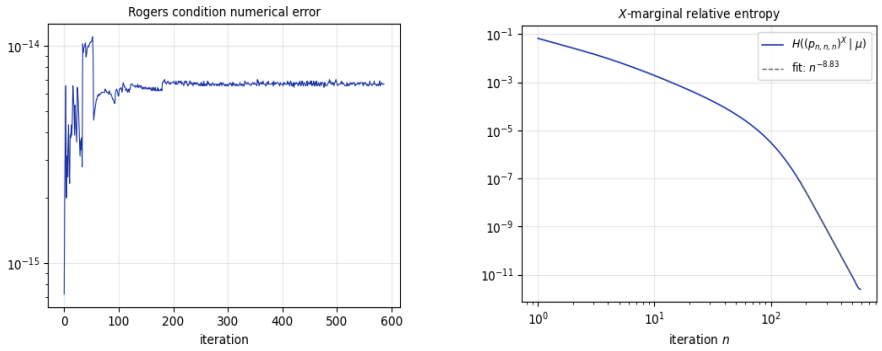


Figure 6: Here we have plotted the error for the Rogers condition for the final $\mu \sim \text{Exp}(1)$ example on the previous page (which is just discretization and root-finding error since theoretically this error is zero for the scheme by construction) and the (entropic) error for the X-marginal. The log-log slope over iterations 177-498 is -8.83, suggesting substantially faster convergence than the proved $O(1/n)$ entropy bound. The code takes about a minute to run on an old MacBook, but we ran to a lower tolerance than is typically necessary in practice.

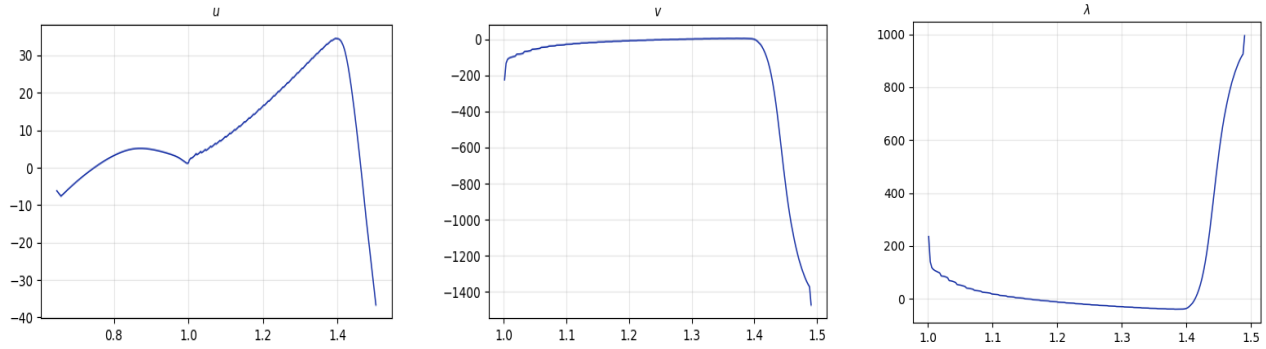


Figure 7: Here we have plotted u, v, λ with μ given by the SVI fit to the 1M EUR-USD smile below with a (made-up) $(\frac{1}{3}, \frac{2}{3})$ AY-Perkins weighting for ν as above, and all other parameters unchanged from Figure 5.

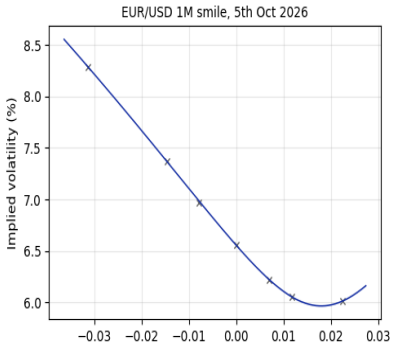


Figure 8: SVI least-squares fit to a 1M EUR-USD smile (the x-axis is log-moneyness here, data taken from table below). The SVI parameters are $(a, b, \sigma, \rho, m) = (0.00016844, 0.00805007, 0.01660353, 0.01493316, 0.01824409)$.

Strike	Call Price	Call Delta	Imp. Vol.
1.0888	0.03571	0.90	8.29%
1.1073	0.01966	0.75	7.37%
1.1148	0.01406	0.65	6.97%
1.1235	0.00852	0.50	6.56%
1.1314	0.00483	0.35	6.22%
1.1369	0.00299	0.25	6.05%
1.1490	0.00094	0.10	6.01%

Table 1: EUR/USD 1M smile on 5th Oct 2026, with spot EUR/USD at 1.1220, data taken from `https://www.investing.com/currencies/eur-usd-options`.

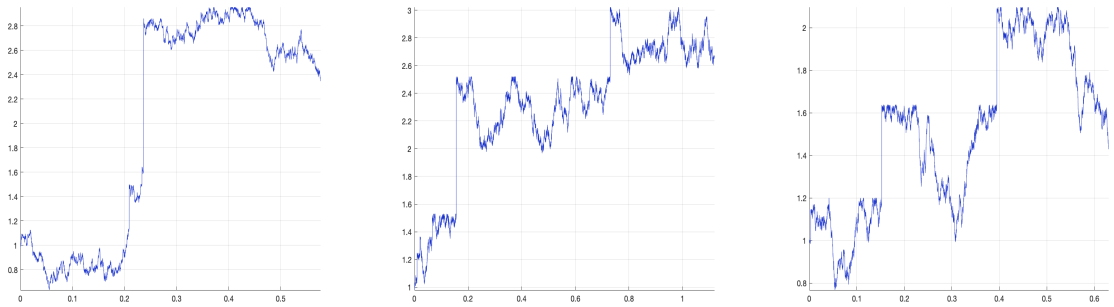


Figure 9: Here we see three simulations of the lost-time Rogers martingale M in Section 4 with target laws $\mu \sim \text{Exp}(1)$ and $\nu([b,\infty)) = \bar{F}_\theta(b) = \exp(-\frac{b(b-1)}{b-1+\theta})$ for $b \geq 1$ and $0 \leq \theta \leq 1$ with $\theta = \frac{1}{2}$ here. At the endpoints, $\bar{F}_1(b) = e^{1-b}$ for $b \geq 1$, while $\bar{F}_0(1) = 1$ and $\bar{F}_0(b) = e^{-b}$ for $b > 1$, and the AY law here is $\text{Exp}(1) + 1$. Note for $\theta = 0$ we require an atom of mass $1 - e^{-1}$ at $b = 1$.

References

[AY79] Azéma, J. and M.Yor, "Une solution simple au problème de Skorokhod", *Séminaire de Probabilités XIII*, Lecture Notes in Mathematics 721, Springer, 90–115, 1979.

[BHR01] Brown, H., D.Hobson, L.C.G. Rogers, "The maximum maximum of a martingale constrained by an intermediate law", *Probability Theory and Related Fields*, 119(4), 558–578, 2001.

[BBBL26] Backhoff,J., M.Beiglböck, G.Bifronte, A.Ley, "Bridging Classical and Martingale Schrödinger Bridges", preprint, arXiv:2604.01299, 2026

[BJ16] Beiglböck, M., Juillet, N., "On a problem of optimal transport under marginal martingale constraints", *Annals of Probability*, 44(1), 42–106, 2016.

[Cox04] Cox, A.M.G., "Skorokhod embeddings: non-centred target distributions, diffusions and minimality", PhD thesis, University of Bath, 2004.

[CO11] Cox, A.M.G. and Obłój, J., "Robust pricing and hedging of double no-touch options", *Finance and Stochastics*, 15(3), 573–605, 2011.

[COT19] Cox, A.M.G., Obłój, J. and Touzi, N., "The Root solution to the multi-marginal embedding problem: an optimal stopping and time-reversal approach", *Probability Theory and Related Fields*, 173(1–2), 211–259, 2019.

[DR15] Duembgen, M., Rogers, L.C.G., "The joint law of the extrema, final value and signature of a stopped random walk", in *In Memoriam Marc Yor – Séminaire de Probabilités XLVII*, Lecture Notes in Mathematics 2137, Springer, 321–338, 2015.

[For26] Forde, M., "Solving the FX cross-smiles problem - rate of convergence for Sinkhorn marginals, and the finite-option case", *Statistics and Probability Letters*, 237, 110805, 2026.

[For19] Forde, M., "Pathwise superhedging for time-dependent barrier options on càdlàg paths – finite or infinite tradeable European, One-Touch, lookback or forward starting options", *Stochastic Processes and their Applications*, 129(3), 799–821, 2019.

[Guy20] Guyon, J., "The Joint S&P 500/VIX Smile Calibration Puzzle Solved", *Risk*, April 2020.

[HLT16] Henry-Labordère, P., Touzi, N., "An explicit martingale version of the one-dimensional Brenier theorem", *Finance and Stochastics*, 20(3), 635–668, 2016.

[HOST16] Henry-Labordère, P., Obłój, J., Spoida, P., Touzi, N., "The maximum maximum of a martingale with given n marginals", *Annals of Applied Probability*, 26(1), 1–44, 2016.

[Hob11] Hobson, D., "The Skorokhod embedding problem and model-independent bounds for option prices", in *Paris–Princeton Lectures on Mathematical Finance 2010*, Lecture Notes in Mathematics 2003, Springer, 267–318, 2011.

[Jac88] Jacka, S.D., "Doob's inequalities revisited: a maximal H^1-embedding", *Stochastic Processes and their Applications*, 29(2), 281–290, 1988.

[JLO23] Joseph, B., G.Loeper and J.Obłój, "The Measure Preserving Martingale Sinkhorn Algorithm", preprint, arXiv:2310.13797, 2023

[Nutz22] Nutz, M., "Introduction to Entropic Optimal Transport", Lecture Notes, 2022.

[Obl04] Obłój, J., "The Skorokhod embedding problem and its offspring", *Probability Surveys*, 1, 321–392, 2004.

[Per86] Perkins, E., "The Cereteli–Davis solution to the H^1-embedding problem and an optimal embedding in Brownian motion", in *Seminar on Stochastic Processes, 1985*, Progress in Probability and Statistics 12, Birkhäuser, Boston, 172–223, 1986.

[Rog93] Rogers, L.C.G., "The joint law of the maximum and terminal value of a martingale", *Probability Theory and Related Fields*, 95(4), 451–466, 1993.

A Proof of Lemma 2.5

Proof. Fix an s for which the hypothesis of the lemma holds, i.e. for which $Z_s(\lambda_2) \in (0,\infty)$ for some finite λ_2, where $Z_s(\lambda) = \int_0^s R(s,x)e^{u(x)+\lambda x}dx$; by assumption this is the case for ν-almost every s. Then, for any other finite λ_1,

$$Z_s(\lambda_1) = \int_0^s e^{(\lambda_1-\lambda_2)x} R(s,x)e^{u(x)+\lambda_2 x}dx.$$

Since $x \in (0,s)$, the factor $e^{(\lambda_1-\lambda_2)x}$ is bounded above and below by strictly positive constants. Hence (by comparison with the previous expression) $0 < Z_s(\lambda_1) < \infty$. Therefore $Z_s(\lambda)$ is positive and finite for every finite λ. Since $x \in (0,s)$, differentiation under the integral sign is justified locally in λ by dominated convergence. Moreover

$$\frac{d}{d\lambda}\log Z_s(\lambda) = \frac{\int_0^s xR(s,x)e^{u(x)+\lambda x}dx}{\int_0^s R(s,x)e^{u(x)+\lambda x}dx} =: m_s(\lambda),$$

which is the mean of x under the probability density $R(s,x)e^{u(x)+\lambda x}/Z_s(\lambda)$ on $(0,s)$, and (using the quotient rule) we can easily check that $m_s'(\lambda)$ is the corresponding variance,

$$m_s'(\lambda) = \int_0^s (x-m_s(\lambda))^2 \frac{R(s,x)e^{u(x)+\lambda x}}{Z_s(\lambda)}dx.$$

By Assumption 2.4, $R(s,x) > 0$ for Lebesgue almost every $x \in (0,s)$. Therefore, since $u(x)$ is finite Lebesgue almost everywhere, we have $\frac{R(s,x)e^{u(x)+\lambda x}}{Z_s(\lambda)} > 0$ for Lebesgue almost every $x \in (0,s)$. The integrand is strictly positive for Lebesgue almost every $x \in (0,s)$ except the single point $x = m_s(\lambda)$. Therefore

$$m_s'(\lambda) > 0$$

for every finite λ. Thus m_s is continuous and strictly increasing. Moreover, since $R(s,x)e^{u(x)} > 0$ for Lebesgue almost every $x \in (0,s)$, for every $\varepsilon \in (0,s)$ we have $\int_0^\varepsilon R(s,x)e^{u(x)}dx > 0$ and $\int_{s-\varepsilon}^s R(s,x)e^{u(x)}dx > 0$. Now fix $\lambda > 0$. Since

$$Z_s(\lambda) = \int_0^s R(s,x)e^{u(x)}e^{\lambda x}dx,$$

and the integrand is non-negative, $Z_s(\lambda) \geq \int_{s-\varepsilon}^s R(s,x)e^{u(x)}e^{\lambda x}dx$. For $x \in (s-\varepsilon, s)$, $e^{\lambda x} \geq e^{\lambda(s-\varepsilon)}$, so

$$Z_s(\lambda) \geq e^{\lambda(s-\varepsilon)} \int_{s-\varepsilon}^s R(s,x)e^{u(x)}dx.$$

Also, since $x < s$, $e^{\lambda x} \leq e^{\lambda s}$, and hence $Z_s(\lambda) \leq e^{\lambda s}\int_0^s R(s,x)e^{u(x)}dx = e^{\lambda s}Z_s(0)$. Therefore

$$e^{\lambda(s-\varepsilon)} \int_{s-\varepsilon}^s R(s,x)e^{u(x)}dx \leq Z_s(\lambda) \leq e^{\lambda s}Z_s(0).$$

Taking logarithms and dividing by λ gives

$$s - \varepsilon + o(1) \leq \frac{\log Z_s(\lambda)}{\lambda} \leq s + o(1)$$

as $\lambda \to \infty$. Since $\varepsilon > 0$ is arbitrary, $\lim_{\lambda\to\infty} \log Z_s(\lambda)/\lambda = s$. Similarly, using $\int_0^\varepsilon R(s,x)e^{u(x)}dx > 0$ and $e^{\lambda x} \geq e^{\lambda\varepsilon}$ for $x \in (0,\varepsilon)$ and $\lambda < 0$, $\lim_{\lambda\to-\infty} \frac{\log Z_s(\lambda)}{\lambda} = 0$. We now use the following elementary fact: for a differentiable convex function f on $\mathbb{R}$, the limits $\lim_{\lambda\to\pm\infty} f'(\lambda)$ exist in $[-\infty,\infty]$ and $\lim_{\lambda\to\pm\infty} f(\lambda)/\lambda = \lim_{\lambda\to\pm\infty} f'(\lambda)$. The function $f = \log Z_s$ is differentiable with $f' = m_s$, and convex because $f'' = m_s' \geq 0$ by the variance formula above. Applying this to $f = \log Z_s$ and combining it with the two limits just computed, we obtain

$$\lim_{\lambda\to-\infty} m_s(\lambda) = 0, \qquad \lim_{\lambda\to\infty} m_s(\lambda) = s.$$

Recall from Eq (12) that $0 < b(s) < s$, so (by the intermediate value theorem) there exists a unique finite $\lambda(s)$ satisfying

$$m_s(\lambda(s)) = b(s).$$

Since $m_s(\lambda)$ is measurable in s for every fixed λ and continuous and strictly increasing in λ for ν-almost every s, the unique root $\lambda(s)$ admits a measurable version.

The final assertion of the lemma was established at the start of the proof, where $Z_s(\lambda)$ was shown to be positive and finite for every finite λ. ■

B The entropy projection onto the S-marginal and the Rogers condition

The following lemma describes what happens when the (λ, v)-update is applied to a probability density q: the resulting density q' satisfies two of the three constraints, namely the S-marginal and the local Rogers conditions, and for any other density p satisfying these two constraints the relative entropies of p and q' with respect to q are related by the Pythagorean identity $H(p|q) = H(p|q') + H(q'|q)$. In particular q' is the density closest to q in relative entropy among those satisfying the two constraints.

Lemma B.1 *Let $D = \{(s,x) : s \geq X_0,\ 0 < x \leq s\}$, and let q be a probability density on D whose S-marginal q^S is positive ν-almost everywhere. For a given measurable function λ which is finite ν-almost everywhere, define*

$$q_\lambda(x|s) = \frac{e^{\lambda(s)x}q(s,x)}{\int_0^s e^{\lambda(s)y}q(s,y)dy}, \qquad x \in (0,s], \text{ for every } s \text{ with } 0 < q^S(s) < \infty,$$

and suppose that λ is such that

$$\int_0^s x q_\lambda(x|s)\,dx = b(s) \qquad \text{for } \nu\text{-almost every } s.$$

Set $q'(s,x) = \nu(s)q_\lambda(x|s)$ for s with $0 < q^S(s) < \infty$ and $q'(s,x) = 0$ otherwise, i.e. q' has S-marginal ν and conditional density $q_\lambda(\cdot|s)$ given $S = s$.

Then q' is a probability density on D, and for any probability density p on D with S-marginal ν, conditional mean $\int_0^s xp(x|s)dx = b(s)$ for ν-almost every s, and $H(p|q) < \infty$, we have $H(p|q) = H(p|q') + H(q'|q)$.

Proof. Step 1: properties of q'. For ν-almost every s, we have $q^S(s) > 0$, and integrating the definition of q' over x gives $\int_0^s q'(s,x)dx = \nu(s)$. Hence q' has S-marginal ν and is a probability density on D. Its conditional density given $S = s$ is $q'(x|s) = e^{\lambda(s)x}q(s,x)/\int_0^s e^{\lambda(s)y}q(s,y)dy$, so the defining equation for λ says precisely that $\int_0^s xq'(x|s)dx = b(s)$ for ν-almost every s: q' has the same conditional mean as p. Moreover, dividing by $q(x|s) = q(s,x)/q^S(s)$,

$$\frac{q'(x|s)}{q(x|s)} = \frac{q^S(s)\, e^{\lambda(s)x}}{\int_0^s e^{\lambda(s)y}q(s,y)dy} \qquad \text{for } \nu\text{-almost every } s, \tag{B-1}$$

whose right-hand side is, for ν-almost every s, positive and finite for every $x \in (0,s)$, since $q^S(s) > 0$, $\lambda(s)$ is finite and $q(s,\cdot)$ is integrable. Hence $q'(\cdot|s)$ and $q(\cdot|s)$ vanish on the same set, i.e. they are mutually absolutely continuous, which we write $q(\cdot|s) \sim q'(\cdot|s)$. Taking logarithms, and noting that neither $q^S(s)$ nor the integral in the denominator depends on x,

$$\log\frac{q'(x|s)}{q(x|s)} = \lambda(s)x + c(s), \qquad c(s) := \log q^S(s) - \log\int_0^s e^{\lambda(s)y}q(s,y)dy, \tag{B-2}$$

so the log-density ratio is an affine function of x.

Step 2: decomposing relative entropy along S. Let P and Q be probability densities on D with $P \ll Q$, and write $P(s,x) = P^S(s)P(x|s)$ and $Q(s,x) = Q^S(s)Q(x|s)$. Then $\log\frac{P(s,x)}{Q(s,x)} = \log\frac{P^S(s)}{Q^S(s)} + \log\frac{P(x|s)}{Q(x|s)}$ for P-almost every (s,x), and integrating against P gives

$$H(P|Q) = H(P^S|Q^S) + \int P^S(s)H(P(\cdot|s)|Q(\cdot|s))ds. \tag{B-3}$$

Step 3: the identity fibrewise. Applying Eq (B-3) with $P = p$, $Q = q$ and using $H(p|q) < \infty$, both terms on the right-hand side are finite; in particular $H(p(\cdot|s)|q(\cdot|s)) < \infty$ for ν-almost every s. Fix such an s. Since $q'(\cdot|s)$ is equivalent to $q(\cdot|s)$ by Step 1, Lemma 1.4(b) in [Nutz22] gives

$$H(p(\cdot|s)|q(\cdot|s)) = H(p(\cdot|s)|q'(\cdot|s)) + \int_0^s \log\frac{q'(x|s)}{q(x|s)}\, p(x|s)dx.$$

By Eq (B-2), for any probability density ρ on $(0,s)$ the integral $\int_0^s \log\frac{q'(x|s)}{q(x|s)}\rho(x)dx = \lambda(s)\int_0^s x\rho(x)dx + c(s)$ depends on ρ only through its mean. Since $p(\cdot|s)$ and $q'(\cdot|s)$ both have mean $b(s)$ (by hypothesis and by Step 1 respectively), the last integral above is unchanged if $p(x|s)$ is replaced by $q'(x|s)$, and equals $H(q'(\cdot|s)|q(\cdot|s))$. Therefore

$$H(p(\cdot|s)|q(\cdot|s)) = H(p(\cdot|s)|q'(\cdot|s)) + H(q'(\cdot|s)|q(\cdot|s)) \qquad \text{for } \nu\text{-almost every } s. \tag{B-4}$$

Step 4: integrating over s. Apply Eq (B-3) to the three pairs (p,q), (p,q') and (q',q); the absolute continuity required holds since $p \ll q$ by hypothesis, $q' \ll q$ by Step 1, and $p \ll q'$ because $p(\cdot|s) \ll q(\cdot|s)$ (from $p \ll q$) and $q(\cdot|s) \sim q'(\cdot|s)$ (Step 1, after Eq (B-1)) for ν-almost every s, and both p and q' have S-marginal ν.

Multiplying Eq (B-4) by $\nu(s)$ and integrating over s gives

$$\int \nu(s)H(p(\cdot|s)|q(\cdot|s))ds = \int \nu(s)H(p(\cdot|s)|q'(\cdot|s))ds + \int \nu(s)H(q'(\cdot|s)|q(\cdot|s))ds. \tag{B-5}$$

We now express each of the three integrals in Eq (B-5) via Eq (B-3), whose first term on the right is the relative entropy of the S-marginals. Here $p^S = \nu$ by hypothesis and $q'^S = \nu$ by Step 1, while q^S is the S-marginal of q. Recall (B-3) states that $H(P|Q) = H(P^S|Q^S) + \int P^S(s)H(P(\cdot|s)|Q(\cdot|s))ds$. Then taking $(P,Q) = (p,q)$, (p,q') and (q',q) respectively gives

$$H(p|q) = H(\nu|q^S) + \int \nu(s)H(p(\cdot|s)|q(\cdot|s))ds, \tag{B-6}$$

$$H(p|q') = \int \nu(s)H(p(\cdot|s)|q'(\cdot|s))ds, \tag{B-7}$$

$$H(q'|q) = H(\nu|q^S) + \int \nu(s)H(q'(\cdot|s)|q(\cdot|s))ds. \tag{B-8}$$

Substituting Eqs (B-6), (B-7) and (B-8) into Eq (B-5), i.e. replacing the three integrals there by $H(p|q) - H(\nu|q^S)$, $H(p|q')$ and $H(q'|q) - H(\nu|q^S)$ respectively, the two terms $H(\nu|q^S)$ cancel and we obtain $H(p|q) = H(p|q') + H(q'|q)$.

■

C Well-definedness of the Sinkhorn scheme

For $n \geq 1$, we regard u_n and v_n as extended-real-valued functions, with $u_n(x) = -\infty$ for $x \notin (0, K)$ and $v_n(s) = -\infty$ for $s \notin (X_0, K)$. The multiplier λ_n is only relevant ν-almost everywhere, and we set $\lambda_n(s) = 0$ whenever $\nu(s) = 0$. Thus $p_{n,n,n}(s, x) = R(s, x)e^{u_n(x)+v_n(s)+\lambda_n(s)x}$ vanishes whenever $x \notin (0, K)$ or $s \notin (X_0, K)$.

Lemma C.1 *Suppose Assumption 2.4 holds and set $u_0 = v_0 = \lambda_0 = 0$. Then every step of the Sinkhorn scheme is well defined. For every $n \geq 0$, $H(p^*|p_{n,n,n}) < \infty$, $H(p^*|p_{n+1,n,n}) < \infty$ and*

$$H(p^*|p_{n,n,n}) - H(p^*|p_{n+1,n,n}) = H(p_{n+1,n,n}|p_{n,n,n}), \tag{C-1}$$

$$H(p^*|p_{n+1,n,n}) - H(p^*|p_{n+1,n+1,n+1}) = H(p_{n+1,n+1,n+1}|p_{n+1,n,n}). \tag{C-2}$$

The functions u_n are finite Lebesgue almost everywhere on $(0, K)$, and λ_n and v_n are finite ν-almost everywhere. Every completed iterate $p_{n,n,n}$ with $n \geq 1$ has S-marginal ν and satisfies the local Rogers condition.

Proof. We proceed by induction. Since $p_{0,0,0} = R$, Assumption 2.4 gives $H(p^*|p_{0,0,0}) < \infty$. Also $u_0 = 0$, $\lambda_0 = 0$ and $v_0 = 0$ are finite, so the finiteness assertions hold at $n = 0$. As the inductive hypothesis, we assume that $H(p^*|p_{n,n,n}) < \infty$, u_n is finite Lebesgue almost everywhere on $(0, K)$, and that λ_n and v_n are finite ν-almost everywhere.

Write $q_n = p_{n,n,n}$ and let q_n^X denote its X-marginal. Since the X-marginals of p^* and q_n are μ and q_n^X, respectively, the DPI gives

$$H(\mu|q_n^X) \leq H(p^*|q_n) < \infty. \tag{C-3}$$

Since $H(\mu|q_n^X) < \infty$, we have $\mu \ll q_n^X$, so the set $\{x \in (0, K) : q_n^X(x) = 0\}$ is μ-null; as $\mu > 0$ Lebesgue almost everywhere on $(0, K)$ by Assumption 2.4, this set is Lebesgue-null, i.e. $q_n^X(x) > 0$ for Lebesgue almost every $x \in (0, K)$. Moreover, since q_n^X is a probability density, $q_n^X(x) < \infty$ for Lebesgue almost every $x \in (0, K)$.

Recall the u-update, Eq (9): $u_{n+1}(x) = \log \mu(x) - \log(\int_x^\infty R(s, x)e^{v_n(s)+\lambda_n(s)x}ds)$. Integrating $q_n(s, x) = R(s, x)e^{u_n(x)+v_n(s)+\lambda_n(s)x}$ over s gives $q_n^X(x) = e^{u_n(x)} \int_x^\infty R(s, x)e^{v_n(s)+\lambda_n(s)x}ds$, so the integral inside the logarithm above equals $e^{-u_n(x)}q_n^X(x)$, and therefore

$$u_{n+1}(x) = \log \mu(x) - \log q_n^X(x) + u_n(x)$$

for Lebesgue almost every $x \in (0, K)$. Hence $p_{n+1,n,n}(s, x)/q_n(s, x) = e^{u_{n+1}(x)-u_n(x)} = \mu(x)/q_n^X(x)$, so $p_{n+1,n,n} \ll q_n$, and $p^* \ll q_n$ since $H(p^*|q_n) < \infty$ by the inductive hypothesis.

Lemma 1.4(b) in [Nutz22] gives

$$H(p^*|q_n) - H(p^*|p_{n+1,n,n}) = \mathbb{E}^{p^*}(\log \frac{p_{n+1,n,n}(S, X)}{q_n(S, X)}) = \mathbb{E}^{p^*}(\log \frac{\mu(X)}{q_n^X(X)}).$$

Since the X-marginal of p^* is μ, $\mathbb{E}^{p^*}(\log \frac{\mu(X)}{q_n^X(X)}) = H(\mu|q_n^X)$, and therefore $H(p^*|q_n) - H(p^*|p_{n+1,n,n}) = H(\mu|q_n^X)$. Since $H(p^*|q_n) < \infty$ and $H(\mu|q_n^X) < \infty$, it follows that $H(p^*|p_{n+1,n,n}) < \infty$.

By Eqs (16) and (C-3), $H(p_{n+1,n,n}|q_n) = H(\mu|q_n^X) < \infty$. Hence

$$H(p^*|q_n) - H(p^*|p_{n+1,n,n}) = H(p_{n+1,n,n}|q_n),$$

which is Eq (C-1).

Since $\mu(x) > 0$ and $0 < q_n^X(x) < \infty$ for Lebesgue almost every $x \in (0, K)$ (shown above), and u_n is finite Lebesgue almost everywhere on $(0, K)$ by the inductive hypothesis, the identity $u_{n+1}(x) = \log \mu(x) - \log q_n^X(x) + u_n(x)$ shows that u_{n+1} is finite Lebesgue almost everywhere on $(0, K)$.

We now apply Lemma 2.5 directly to u_{n+1}. Since $p_{n+1,n,n}(s, x) = R(s, x)e^{u_{n+1}(x)+v_n(s)+\lambda_n(s)x}$, its S-marginal is

$$(p_{n+1,n,n})^S(s) = e^{v_n(s)} \int_0^s R(s, x)e^{u_{n+1}(x)+\lambda_n(s)x}dx.$$

Since $p_{n+1,n,n}$ is a probability density, $(p_{n+1,n,n})^S$ is integrable, and hence finite for Lebesgue almost every s. As shown above, $H(p^*|p_{n+1,n,n}) < \infty$. Since the S-marginal of p^* is ν, the DPI gives $H(\nu|(p_{n+1,n,n})^S) \leq H(p^*|p_{n+1,n,n}) < \infty$, hence $\nu \ll (p_{n+1,n,n})^S$, i.e. $(p_{n+1,n,n})^S(s) > 0$ for ν-almost every s.

We have already shown that $0 < (p_{n+1,n,n})^S(s) < \infty$ for ν-almost every s. Moreover, v_n is finite ν-almost everywhere by the inductive hypothesis, and hence $0 < e^{-v_n(s)} < \infty$ for ν-almost every s. Therefore

$$0 < \int_0^s R(s, x)e^{u_{n+1}(x)+\lambda_n(s)x}dx < \infty$$

for ν-almost every s. Hence the hypotheses of Lemma 2.5 hold, so the λ-update is well defined and there exists a (ν a.e.) unique finite measurable function $\lambda_{n+1}(s)$ satisfying

$$\frac{\int_0^s xR(s, x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x}dx}{\int_0^s R(s, x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x}dx} = b(s).$$

By Assumption 2.4 and the fact that ν is a probability density, $0 < \nu(s) < \infty$ for ν-almost every s. Now recall Eq (11):

$$v_{n+1}(s) = \log \nu(s) - \log(\int_0^s R(s,x) e^{u_{n+1}(x)+\lambda_{n+1}(s)x} dx).$$

Then $\log \nu(s)$ is finite for ν-almost every s, and the second logarithm in Eq (11) is finite for ν-almost every s by Lemma 2.5 applied with $u = u_{n+1}$ and $\lambda = \lambda_{n+1}(s)$, so the v-update is well defined and v_{n+1} is finite ν-almost everywhere.

We now apply Lemma B.1 with $q = p_{n+1,n,n}$ and $\lambda(s) = \lambda_{n+1}(s) - \lambda_n(s)$. Its hypotheses hold. First, the S-marginal of q is positive ν-almost everywhere (shown above). Second, by definition $q_\lambda(x|s) \propto e^{\lambda(s)x} q(s,x)$; with $\lambda = \lambda_{n+1} - \lambda_n$ and $q = p_{n+1,n,n} = R\, e^{u_{n+1}(x)+v_n(s)+\lambda_n(s)x}$ this is

$$e^{(\lambda_{n+1}(s)-\lambda_n(s))x}\, R(s,x) e^{u_{n+1}(x)+v_n(s)+\lambda_n(s)x} = e^{v_n(s)}\, R(s,x) e^{u_{n+1}(x)+\lambda_{n+1}(s)x}.$$

Hence $q_\lambda(x|s) = R(s,x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x} / \int_0^s R(s,y)e^{u_{n+1}(y)+\lambda_{n+1}(s)y} dy$, whose mean $\int_0^s x q_\lambda(x|s) dx$ is the left-hand side of Eq (10), hence equal to $b(s)$. Finally, by the definition of q' in Lemma B.1,

$$\begin{aligned} q'(s,x) = \nu(s)\, q_\lambda(x|s) = \frac{\nu(s)}{\int_0^s R(s,y) e^{u_{n+1}(y)+\lambda_{n+1}(s)y} dy} R(s,x) e^{u_{n+1}(x)+\lambda_{n+1}(s)x} &= e^{v_{n+1}(s)} R(s,x) e^{u_{n+1}(x)+\lambda_{n+1}(s)x} \\ &= p_{n+1,n+1,n+1}(s,x), \end{aligned}$$

where the third equality is Eq (11).

p^* has S-marginal ν, conditional mean b and $H(p^*|p_{n+1,n,n}) < \infty$ (shown above), so Lemma B.1 gives

$$H(p^*|p_{n+1,n,n}) = H(p^*|p_{n+1,n+1,n+1}) + H(p_{n+1,n+1,n+1}|p_{n+1,n,n}) \tag{C-4}$$

which is Eq (C-2). Since the left-hand side is finite and both terms on the right are nonnegative,

$$H(p^*|p_{n+1,n+1,n+1}) < \infty.$$

Finally, $(p_{n+1,n+1,n+1})^S = \nu$ by Lemma B.1, since $p_{n+1,n+1,n+1}$ is the density q' there, and by Eq (10)

$$\int_0^s x p_{n+1,n+1,n+1}(s,x) dx = b(s)\nu(s) = r(s),$$

so the completed iterate satisfies the local Rogers condition. This completes the induction. ∎

D The Rogers construction

For completeness we sketch the construction in Section 3 of [Rog93] of a continuous martingale with a prescribed joint law of (S_T, X_T), in the case relevant here where the joint law has a density. It is convenient to work with $Y = S - X$ and to write $\bar\mu(b,y)$ for the joint density of (S_T, Y_T) on $[x_0,\infty)\times(0,\infty)$, so that $\bar\mu(b,y) = \pi(b, b-y)$ in the notation of Theorem 1.1.

Let W be a Brownian motion started at x_0 with running maximum $\bar W_t = \max_{u\le t} W_u$, and let $\{Z_b\}_{b\ge x_0}$ be a family of random variables, independent of W and of each other, with Z_b having density $f(\cdot\,;b)$ on $(0,\infty)$. Define $\tau = \inf\{t \ge 0 :\ \bar W_t - W_t \ge Z_{\bar W_t}\}$. The rate of excursions with drawdown $\ge y$ is $\frac{1}{y}$, so the excursions that reach Z_b arrive at rate $\int_0^\infty y^{-1} f(y;b) dy$. Then

$$\begin{aligned} \mathbb{P}(\bar W_\tau > b) &= \exp[-\int_{x_0}^b \int_0^\infty \frac{1}{y} f(y;s) dy ds], \\ \mathbb{P}(\bar W_\tau \in db,\ \bar W_\tau - W_\tau \in dy) &= \mathbb{P}(\bar W_\tau \ge b)\, \frac{1}{y} f(y;b) dy db. \end{aligned} \tag{D-1}$$

Now let $\bar\mu(b,y)$ be the target density and $\bar\rho(b) = \int_b^\infty \int_0^\infty \bar\mu(s,u)\, du\, ds = \mathbb{P}(S_T > b)$. Matching (D-1) with $\bar\mu$ requires $\mathbb{P}(\bar W_\tau \ge b) = \bar\rho(b)$ and

$$f(y;b) = \frac{y\bar\mu(b,y)}{\bar\rho(b)}.$$

This definition is used only for levels b with $\bar\rho(b) > 0$; at other levels the mark distribution may be chosen arbitrarily.

Since $\nu(b) = \int_0^\infty \bar\mu(b,y) dy$, we have $\int_0^\infty y^{-1} f(y;b) dy = \nu(b)/\bar\rho(b) = -\bar\rho'(b)/\bar\rho(b)$. Hence, replacing the inner integral in the first identity above and using $\bar\rho(x_0) = 1$ gives

$$\mathbb{P}(\bar W_\tau > b) = \exp(\int_{x_0}^b \frac{\bar\rho'(s)}{\bar\rho(s)} ds) = \frac{\bar\rho(b)}{\bar\rho(x_0)} = \bar\rho(b). \tag{D-2}$$

This is a legitimate choice only if $f(\cdot\,;b)$ is a probability density in y for each b, i.e.

$$\int_0^\infty y\bar{\mu}(b,y)dy = \bar{\rho}(b) \qquad \text{for almost every } b \geq x_0. \tag{D-3}$$

Substituting $\bar{\mu}(b,y) = \pi(b, b-y)$, the left-hand side of (D-3) is $\int_{-\infty}^{b}(b-x)\pi(b,x)dx = b\nu(b) - \eta(b)$ and $\bar{\rho}(b) = \bar{F}_\nu(b)$, so (D-3) is exactly the local Rogers condition (1). Since $\bar{\rho}(b) \to 0$ as $b \to \infty$, the stopping time τ is almost surely finite. The constructed joint law is then the prescribed law π, and its integrability and mean condition from Theorem 1.1 imply that $(W_{t\wedge\tau})_{t\geq 0}$ is uniformly integrable, by Lemma 2.3 of [Rog93], applied after subtracting x_0.

E Proof of Proposition 2.3

To see why $b(\cdot)$ is strictly increasing, recall that $\bar{F}'_\nu = -\nu$. First, $\frac{d}{ds}\frac{1}{\bar{F}_\nu(s)} = -\frac{\bar{F}'_\nu(s)}{\bar{F}_\nu(s)^2} = \frac{\nu(s)}{\bar{F}_\nu(s)^2}$, so $1/\bar{F}_\nu$ is strictly convex if and only if $s \mapsto \nu(s)/\bar{F}_\nu(s)^2$ is strictly increasing, in which case (taking logs) its derivative is non-negative and vanishes on no interval:

$$\frac{d}{ds}\log\frac{\nu(s)}{\bar{F}_\nu(s)^2} = \frac{\nu'(s)}{\nu(s)} - 2\frac{\bar{F}'_\nu(s)}{\bar{F}_\nu(s)} = \frac{\nu'(s)}{\nu(s)} + \frac{2\nu(s)}{\bar{F}_\nu(s)} \geq 0.$$

Second, by the quotient rule,

$$b'(s) = 1 - \frac{\bar{F}'_\nu(s)\nu(s) - \bar{F}_\nu(s)\nu'(s)}{\nu(s)^2} = 1 + \frac{\nu(s)^2 + \bar{F}_\nu(s)\nu'(s)}{\nu(s)^2} = 2 + \frac{\bar{F}_\nu(s)\,\nu'(s)}{\nu(s)^2}.$$

Multiplying by $\nu(s)/\bar{F}_\nu(s) > 0$, $b'(s)$ has the sign of $\frac{2\nu(s)}{\bar{F}_\nu(s)} + \frac{\nu'(s)}{\nu(s)}$, the quantity just obtained. Hence $b' \geq 0$, with $b' > 0$ outside a set with empty interior, so b is strictly increasing.

We break the remainder of the proof into two parts:

(i) Since b is strictly increasing on the support of ν, S_T and $Z = b(S_T)$ generate the same σ-algebra, so the local Rogers condition (1) can be rewritten as $\mathbb{E}(X_T|S_T) = b(S_T) \iff \mathbb{E}(X_T|Z) = Z$, i.e. a joint law of (S_T, X_T) with marginals ν and μ satisfies the local Rogers condition if and only if the law of (Z, X_T) is a martingale coupling of ν_Z and μ. By Strassen's theorem, such a martingale coupling m exists if and only if $\mu \succeq_{\text{cx}} \nu_Z$.

Given such an m, let π be the law of (S_T, X_T) when $(Z, X_T) \sim m$ and $S_T = b^{-1}(Z)$, i.e. $\pi(ds, dx) = \nu(ds)\, m_{b(s)}(dx)$. By the equivalence above, π has marginals ν and μ and satisfies the local Rogers condition, and the remaining conditions of Theorem 1.1 (which concern only the marginals) hold by Assumption 2.1. Hence π is admissible if and only if it also satisfies the support condition $x \leq s$, i.e. if and only if $m_{b(s)}$ charges only $(-\infty, s]$ for ν-almost every s, which is the stated condition on m_z with $z = b(s)$.

(ii) By [BJ16], under $\mu \succeq_{\text{cx}} \nu_Z$ the left-curtain coupling of (ν_Z, μ) exists and by [HLT16] it is the two-point kernel $m_z = q\,\delta_{T_u(z)} + (1-q)\,\delta_{T_d(z)}$, $q = (z - T_d(z))/(T_u(z) - T_d(z))$. The single-crossing condition in the statement ensures that this coupling has no identity part (cf. [HLT16]), so for ν_Z-almost every z the branches $T_d(z) \leq z \leq T_u(z)$ are characterised by the two identities

$$\mu([T_d(z), T_u(z)]) = \nu_Z((b(x_0), z]), \qquad \int_{T_d(z)}^{T_u(z)} x\mu(dx) = \int_{b(x_0)}^{z} z'\nu_Z(dz').$$

Substituting $z = b(s)$, $x_\pm(s) = T_{u/d}(b(s))$, and using $\nu_Z((b(x_0), b(s)]) = \nu((x_0, s]) = 1 - \bar{F}_\nu(s)$ and $\int_{b(x_0)}^{b(s)} z'\nu_Z(dz') = \int_{x_0}^{s} b(u)\nu(u)du$, these are the two displayed equations in the statement.